

Universality of 1-Variable Automata is Decidable¹

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Regular Languages over Finite Alphabets

Finite-State Automata

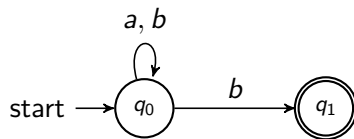


Figure: An FSA recognizing all words that end in the letter b .

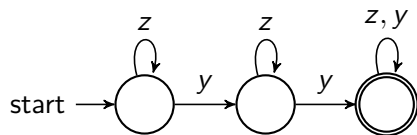
- ▶ the original
- ▶ only works for finite alphabets
- ▶ important decision problems decidable. Of interest: universality, i.e., does the automaton accept every word?

Automata over Infinite Alphabets

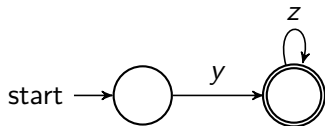
	Finite-State Automata	Register Automata	1-Variable RA	Variable Automata	1-Variable VA
Union, Intersection Closed	✓	✓	✗	✓	✗
Complementation Closed	✓	✗	✗	✗	✗
Universality Decidable	✓	✗	✓	✗	?

Variable Automata

Variable Automata
(Grumberg et al. 2010)



(a) A variable automaton V_1 recognizing all words where some letter occurs twice.



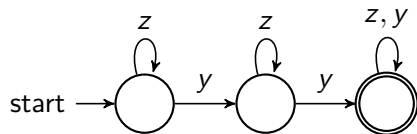
(b) A VA V_2 accepting all words where the first letter is unique.

Figure: Examples of VA.

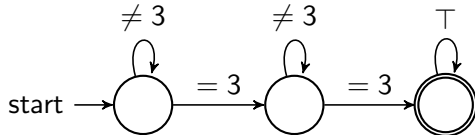
- ▶ bound variables $Y = \{y_1, y_2, \dots, y_k\}$, correspond to letters of alphabet D
- ▶ free variable z represents every letter not stored in a bound variable
- ▶ a word is accepted if there exists a variable assignment $\mu : Y \rightarrow D$ such that it matches an accepting run.

Variable Automata

Variable Automata
(Grumberg et al. 2010)



(a) A variable automaton V_1 recognizing all words where some letter occurs twice.



(b) Visualizing parameter assignment $\mu(y) = 3$.

Figure: Examples of VA.

- ▶ Let $D = \mathbb{N}$
- ▶ $(3, 2, 3)$ is accepted with variable assignment $\mu(y) = 3$,
- ▶ $(1, 2, 1, 2)$ is accepted with $\mu(y) = 1$ or $\mu(y) = 2$,
- ▶ $(1, 2, 3)$ is rejected (no accepting run)

Universality Problem: Running Example

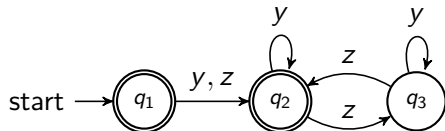


Figure: An example VA V .

We investigate the universality problem for the 1-VA fragment:

- ▶ one bound variable
- ▶ deterministic per assignment (i.e., one outgoing transition for each variable and state)

Theorem

The universality problem is decidable for 1-VA.

Idea: Positions

Definition

A word's *position* is the multiset of all potential terminating states.

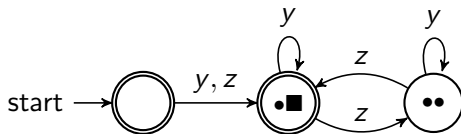


Figure: The position of $w = abacb$ in V , envisioned with tokens.

- ▶ one element for each distinct letter of the word
- ▶ all runs where $\mu(y)$ is not a letter of the word can be grouped into one
- ▶ a 1-VA is universal if all reachable positions contain at least one accepting state.

Observations About Positions

- ▶ d distinct letters $\rightarrow$ position has $d + 1$ elements
- ▶ different words can have the same position (e.g., abc and $abacb$)
- ▶ isomorphic words (e.g., aab and cdd) have the same position
- ▶ positions form a transition system: from word w with position p , we can compute the position of successor $w\alpha$ for any letter $\alpha \in D$
- ▶ the position of a successor does not depend on a concrete letter, but on the terminating state when using the relevant parameter assignment

How to draw an owl

1.



2.



1. Draw some circles

2. Draw the rest of the fucking owl

The Rest of the Owl

- ▶ No entries below 0, shifting of entries can be modelled using matrices
- ▶ the transition system is an *affine Vector Addition System*
- ▶ problem: reachability in affine VAS is undecidable!
- ▶ observation: effect of matrix operations is limited and can be accounted for by introducing states
- ▶ model as intersection of VAS and semilinear set. This is decidable²³!

²it is also Ackermann complete

³that does not mean the universality problem for 1-VA is Ackermann complete, this is only an upper bound

In Conclusion

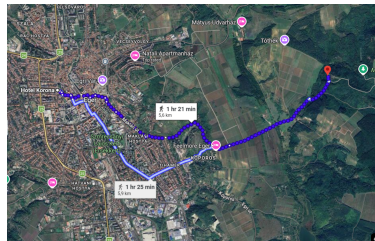
- ▶ Universality of 1-VA is decidable
- ▶ not very happy with my proof yet



[https://alber.me/
infinite-alphabets/positions.html](https://alber.me/infinite-alphabets/positions.html)



Hike?



Kaptárkő 15 instanTrail

